

Reconstruction and Modeling for Physical World Interactions

Zhilu Zhang (张志路)

Singapore Management University

Physical Interactions Shape Our World



**People and robots
change the world
through interactions**

Reconstruction, Prediction, and Action

Reconstruct

What is happening?



Recover appearance,
geometry, semantics...



Predict

What would happen?



Model future states
under candidate actions

Act

Which action achieves
the goal?



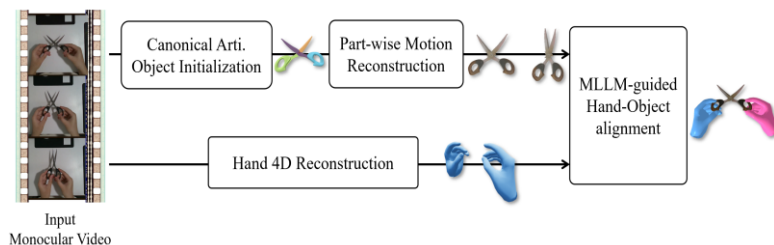
Plan and execute
an action toward a goal

New observations update the state

Content

ArtHOI (CVPR 2026, Highlight)

Interaction reconstruction

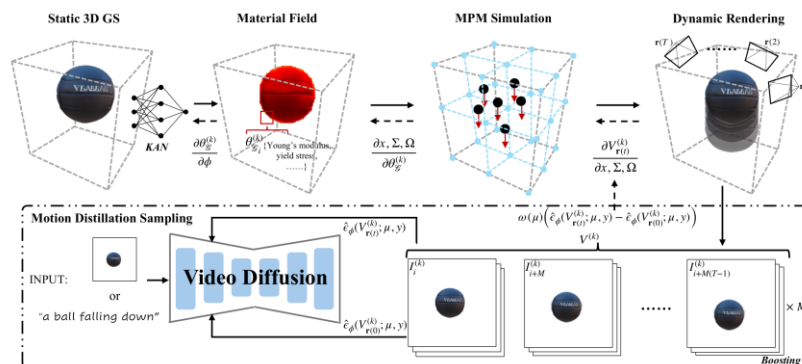


Reconstruct 4D hand-articulated-object interactions

How did the interaction occur?

DreamPhysics (AAAI 2025)

Physical response modeling

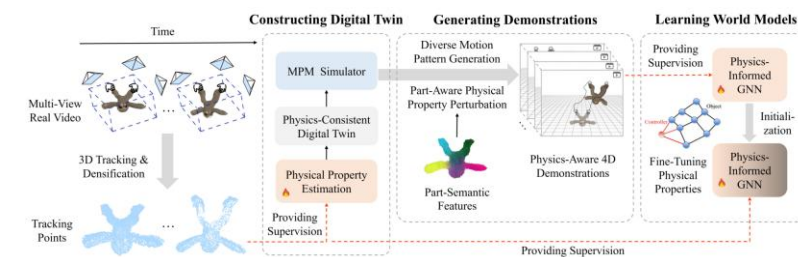


Learn material fields with video diffusion priors

How can the object respond?

PhysWorld (Under Review)

Predictive world modeling



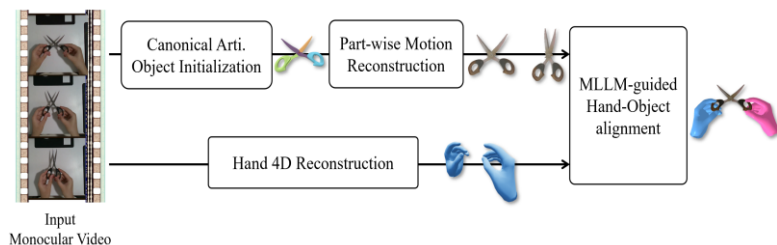
Synthesize physical demonstrations and learn efficient dynamics model of deformable objects

What follows a new action?

Content

ArtHOI (CVPR 2026, Highlight)

Interaction reconstruction

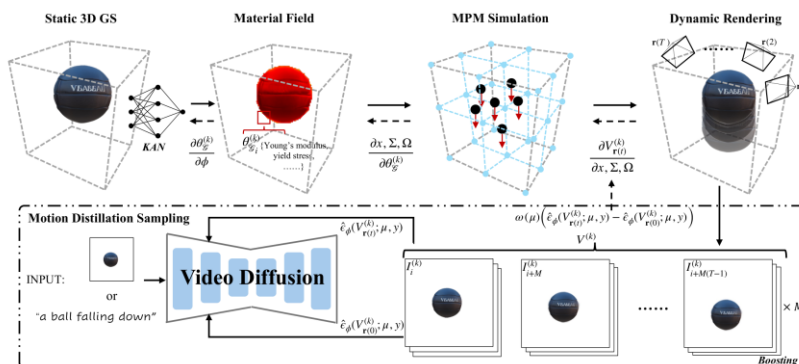


Reconstruct 4D hand-articulated-object interactions

How did the interaction occur?

DreamPhysics (AAAI 2025)

Physical response modeling

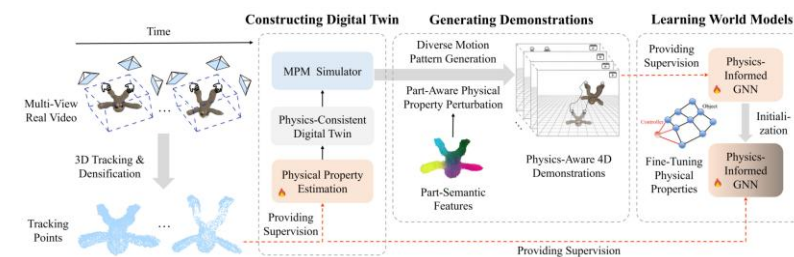


Learn material fields with video diffusion priors

How can the object respond?

PhysWorld (Under Review)

Predictive world modeling



Synthesize physical demonstrations and learn efficient dynamics model of deformable objects

What follows a new action?

Hand-Object Interactions (HOI) reconstruction

- Obtain a physically plausible 3D representation of hands, objects, and their interplay from visual observations.
- It can be applied to robot manipulation and human behavior analysis.



Previous Works

- Handle only **rigid objects**
- Assum **fixed hand-object contact**

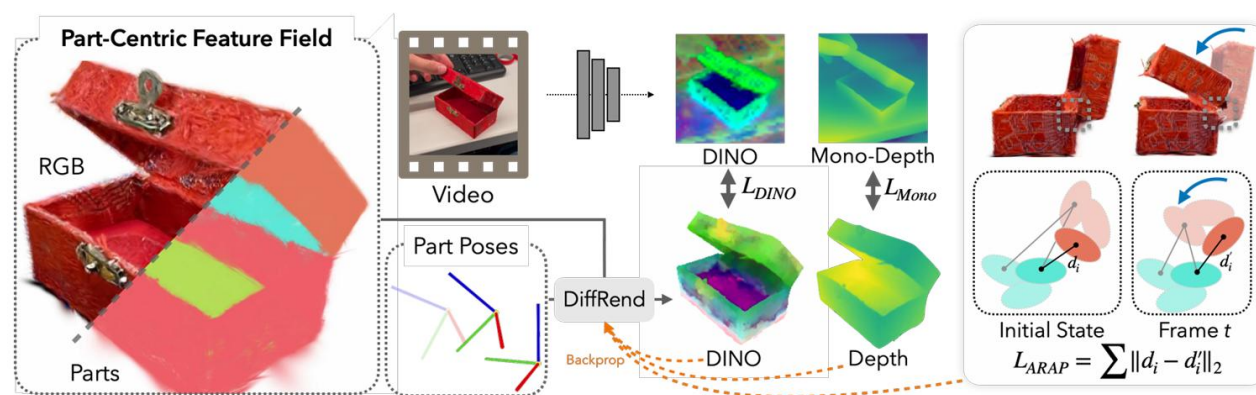
EasyHOI: Unleashing the Power of Large Models for Reconstructing Hand-Object Interactions in the Wild



EasyHOI. CVPR 2025

- Require a **3D scan** of an object.
- Lack modeling of **the interacting hand**

Robot See Robot Do: Imitating Articulated Object Manipulation with Monocular 4D Reconstruction



RSRD. CoRL 2024

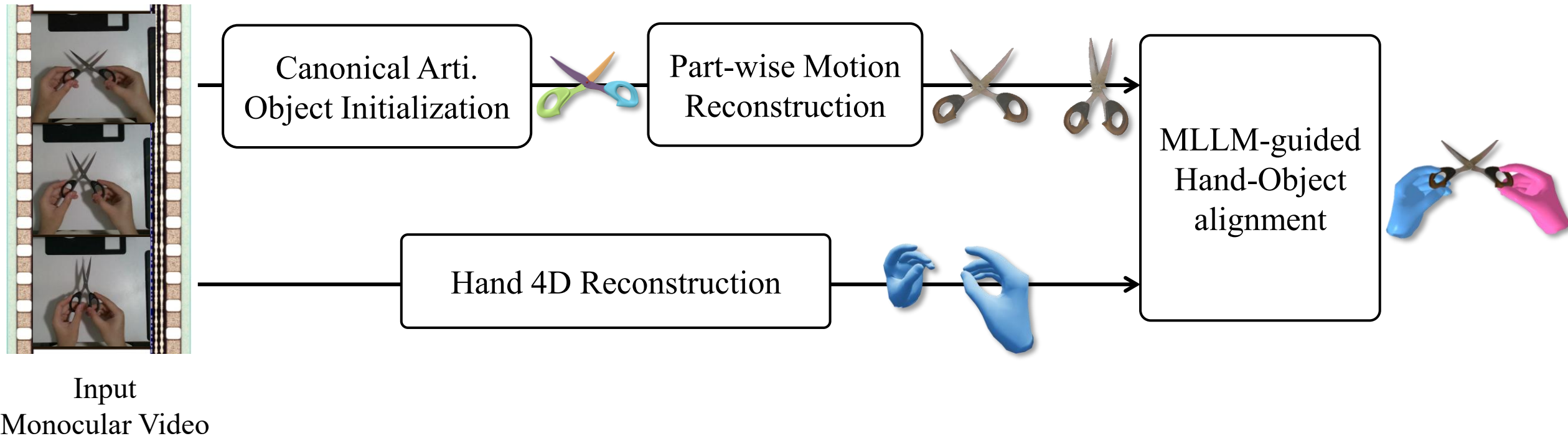
Our goal and motivation

- Goal
 - Reconstruct 4D hand-articulated-object interactions from a monocular video
- Method Motivation
 - Humans can effortlessly perceive such complex interactions, a capability that stems from accumulated knowledge and experience.
 - Machines can leverage the rich priors of various foundation models.



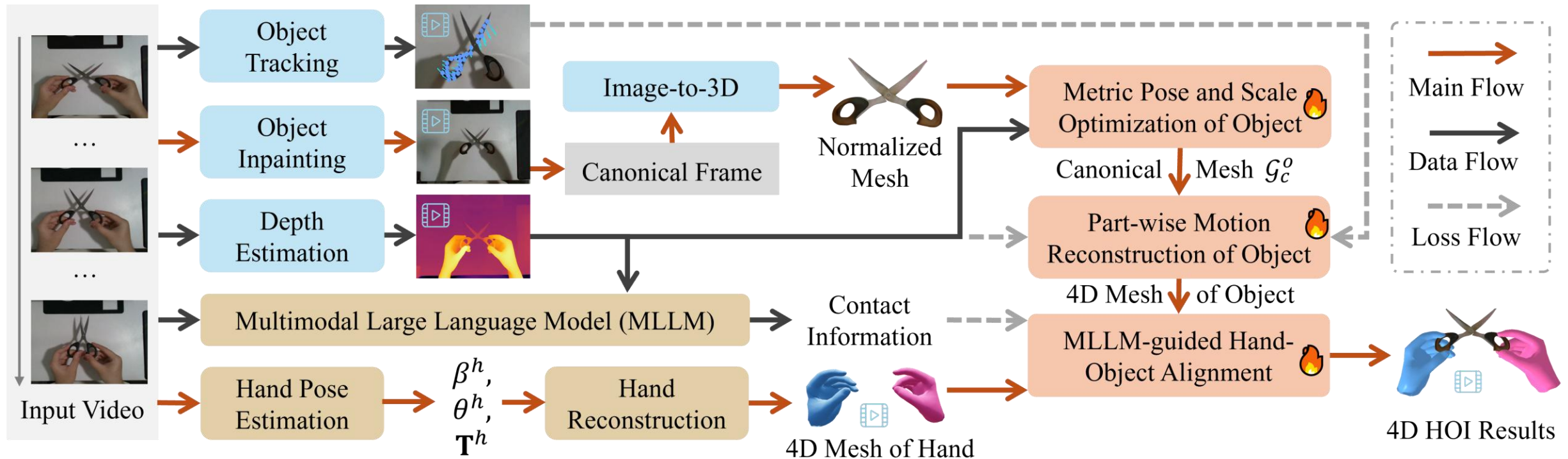
Pipeline Overview

- First, perform part-wise motion reconstruction and 4D reconstruction of the hand.
- Then, perform hand-object alignment.



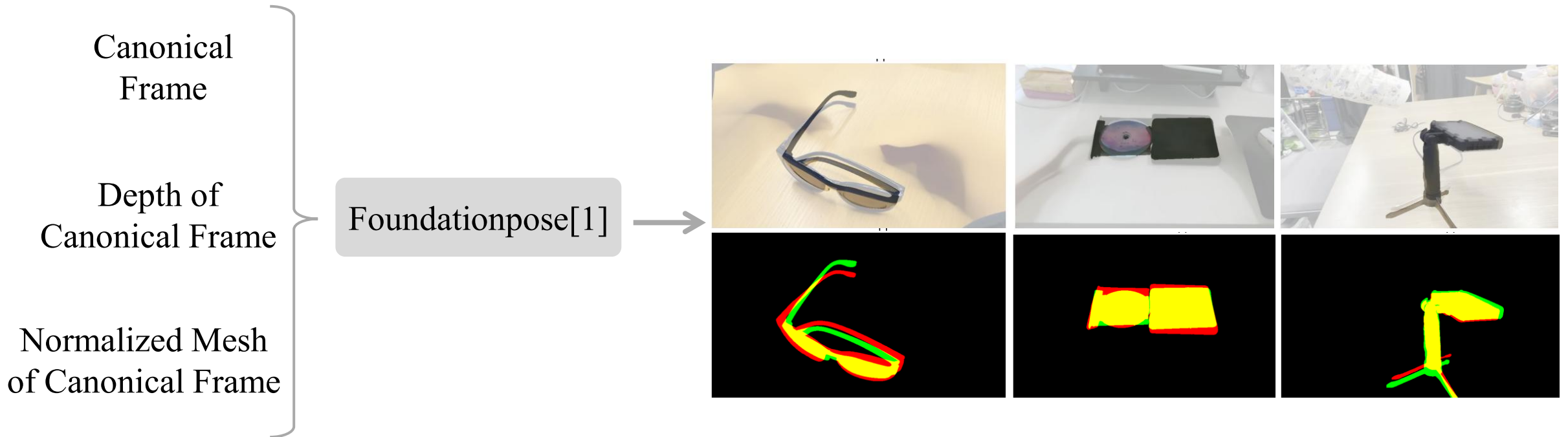
Pipeline Overview

- Integrates priors from multiple foundation models
- Optimize the inconsistency and mismatch problems while leveraging priors.



Metric Pose and Scale Optimization

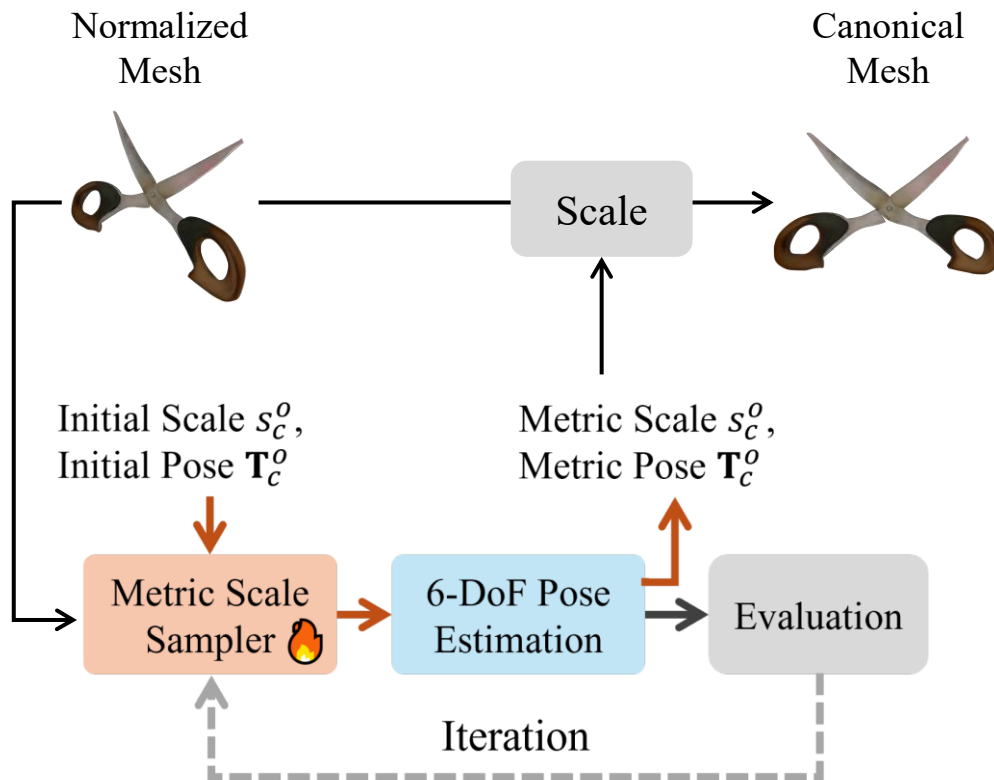
- Recover metric scale and 6-DoF pose at world space



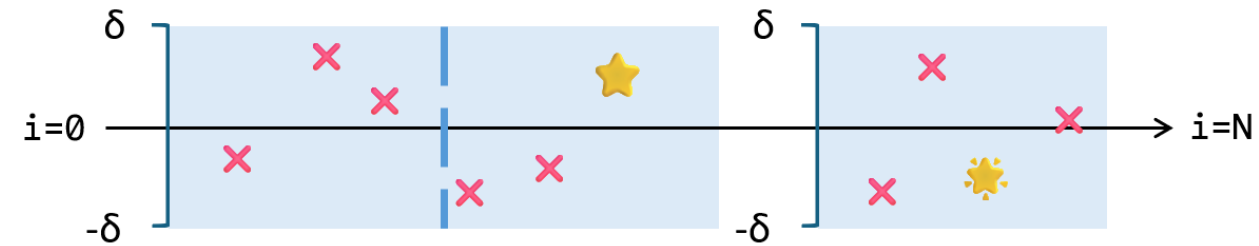
Inconsistent, Inaccurate Priors → Low-quality Initialization!

Metric Pose and Scale Optimization

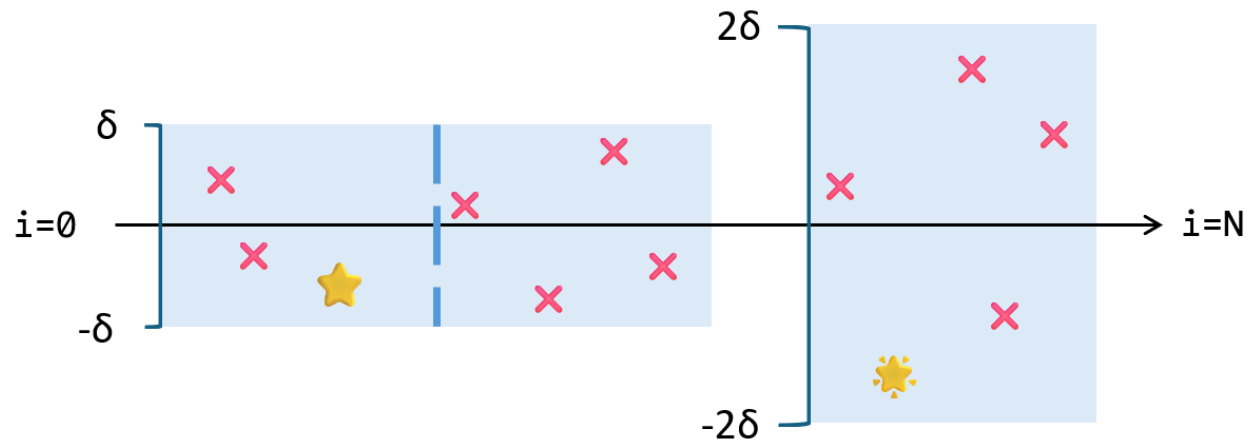
- Adaptive Sampling Refinement (ASR)



Best found late: current range is still producing better samples, so ASR **keeps** current range:



Best found early: current range is likely saturated, so ASR **expands** the range for exploration:



Part-wise Motion Reconstruction

Tracks $Z_0^{P_k} \dots Z_N^{P_k}$ Visibilities $V_0^{P_k} \dots V_N^{P_k}$



$$\mathcal{L}_{\text{track}} = \sum_{j \in \mathbb{S}} \sum_{q \in \mathbb{W}_{i,j}^k} \left\| \mathbf{z}_{j,q}^k - (\mathbf{T}_i^{p_k})^{-1} \mathbf{T}_j^{p_k} \mathbf{z}_{i,q}^k \right\|,$$

where $\mathbb{W}_{i,j}^k = \{q \mid v_{i,q}^k = 1 \wedge v_{j,q}^k = 1\}$



SE(3) Part Poses

Predicted Rendered



Predicted Mask



Ground Truth RGB



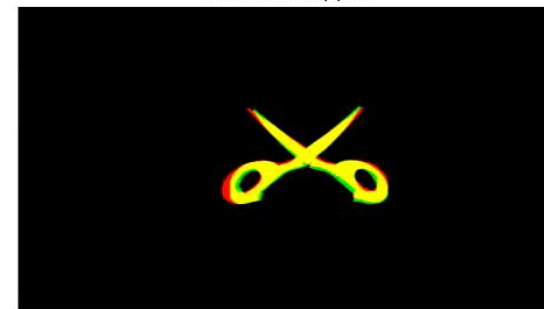
Ground Truth Mask



RGB Overlapped

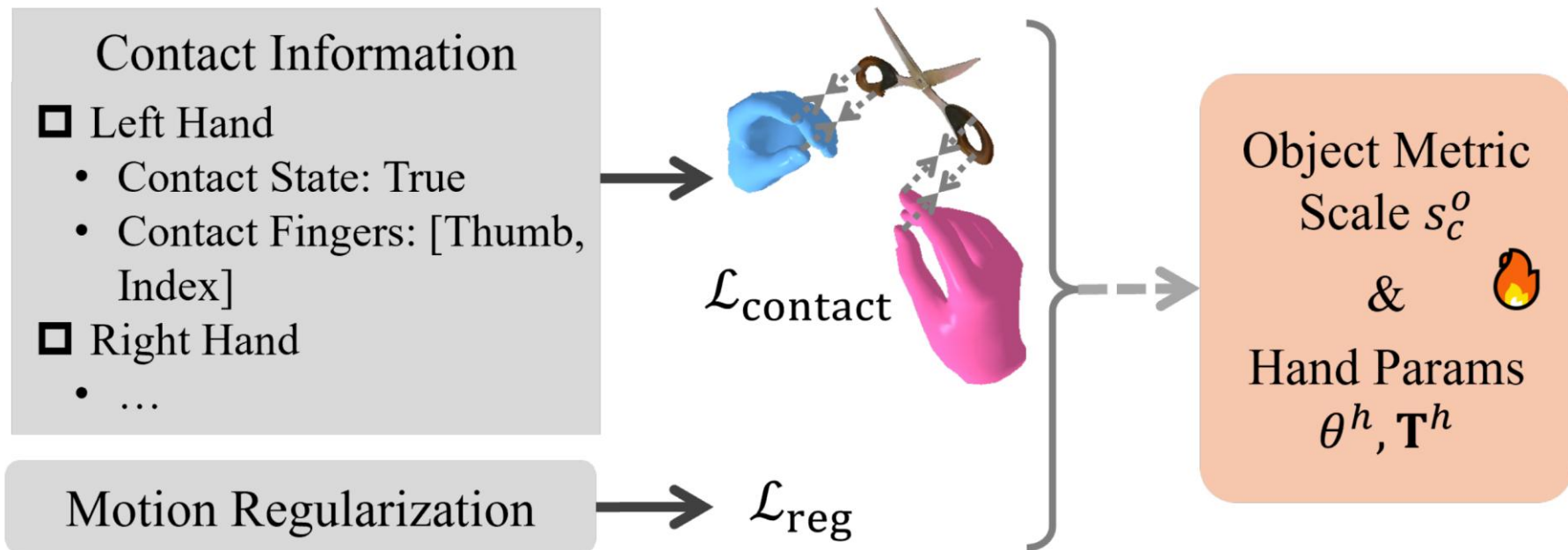


Mask Overlapped



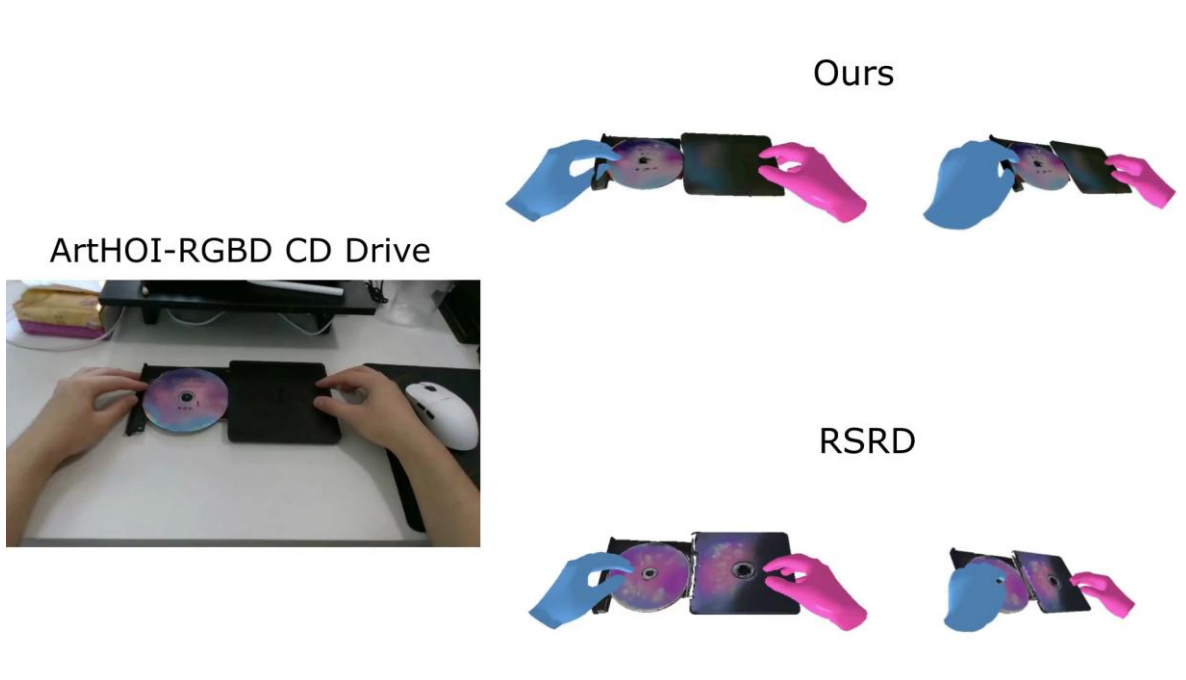
MLLM-guided Hand-Object Alignment

- Ask the MLLM to infer frame-wise contact information, including hand laterality, binary contact state, and contacting fingers.
- Take the contact information as frame-wise constraints to guide 4D hand-object interaction alignment.



Experimental Results

- Reconstruct articulated objects and interacting hands together.



Comparison with RSRD[1] on the ArtHOI-RGBD dataset.

Experimental Results

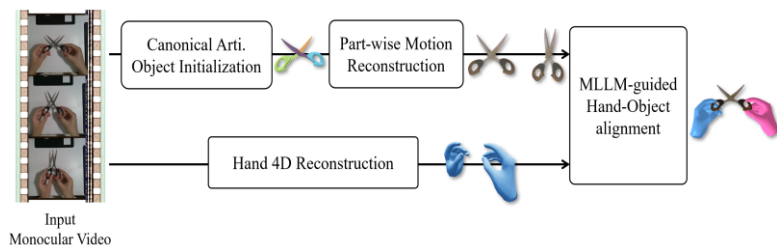
- Reconstruction from everyday videos of glasses and a toy blaster.



Content

ArtHOI (CVPR 2026, Highlight)

Interaction reconstruction

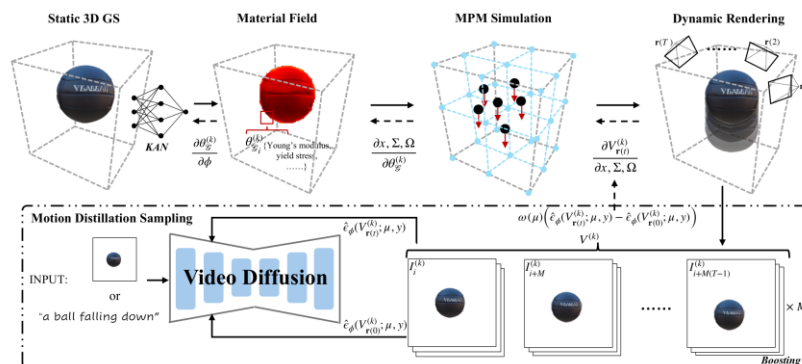


Reconstruct 4D hand-articulated-object interactions

How did the interaction occur?

DreamPhysics (AAAI 2025)

Physical response modeling

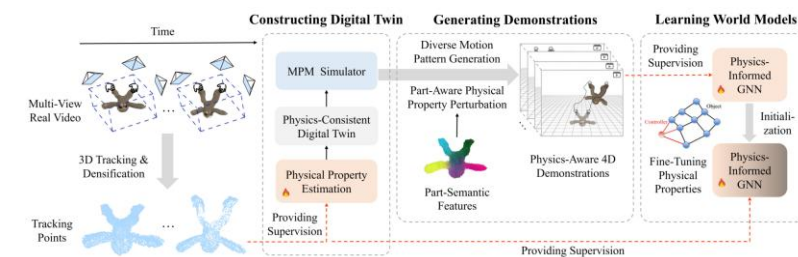


Learn material fields with video diffusion priors

How can the object respond?

PhysWorld (Under Review)

Predictive world modeling

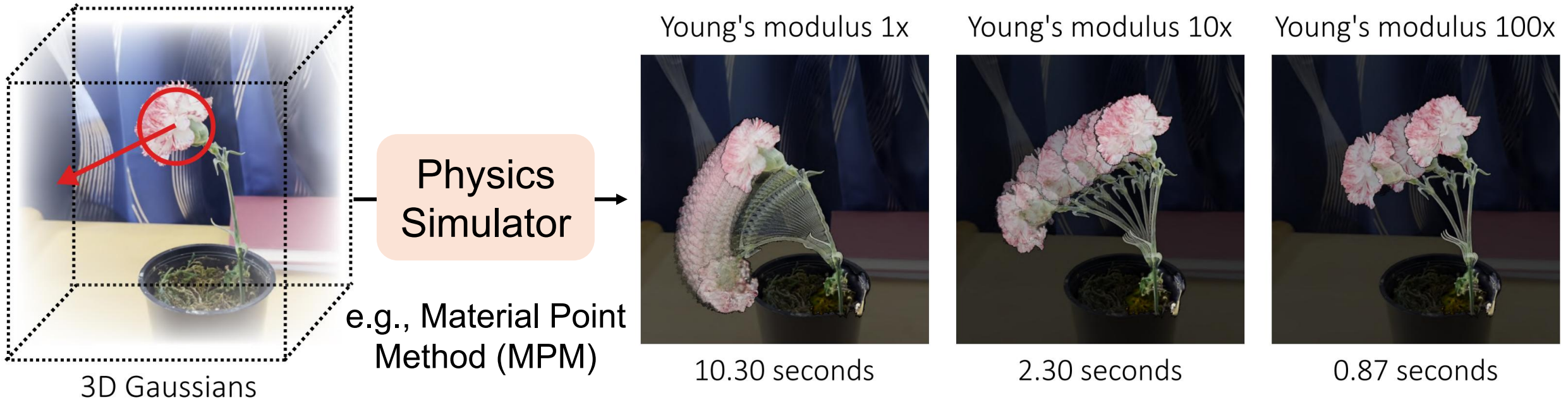


Synthesize physical demonstrations and learn efficient dynamics model of deformable objects

What follows a new action?

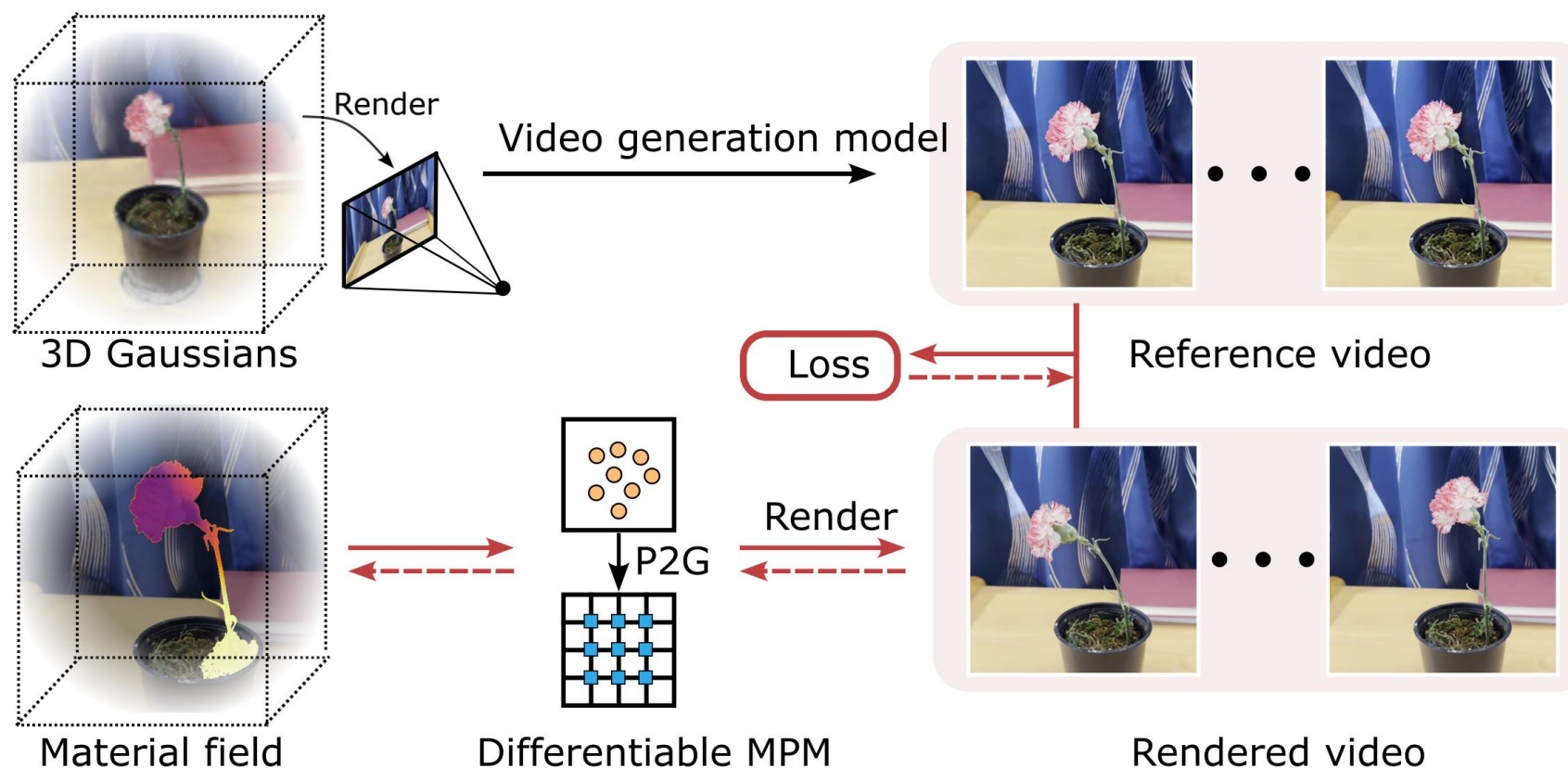
Physical Properties

- Physical properties determine how a 3D object moves.



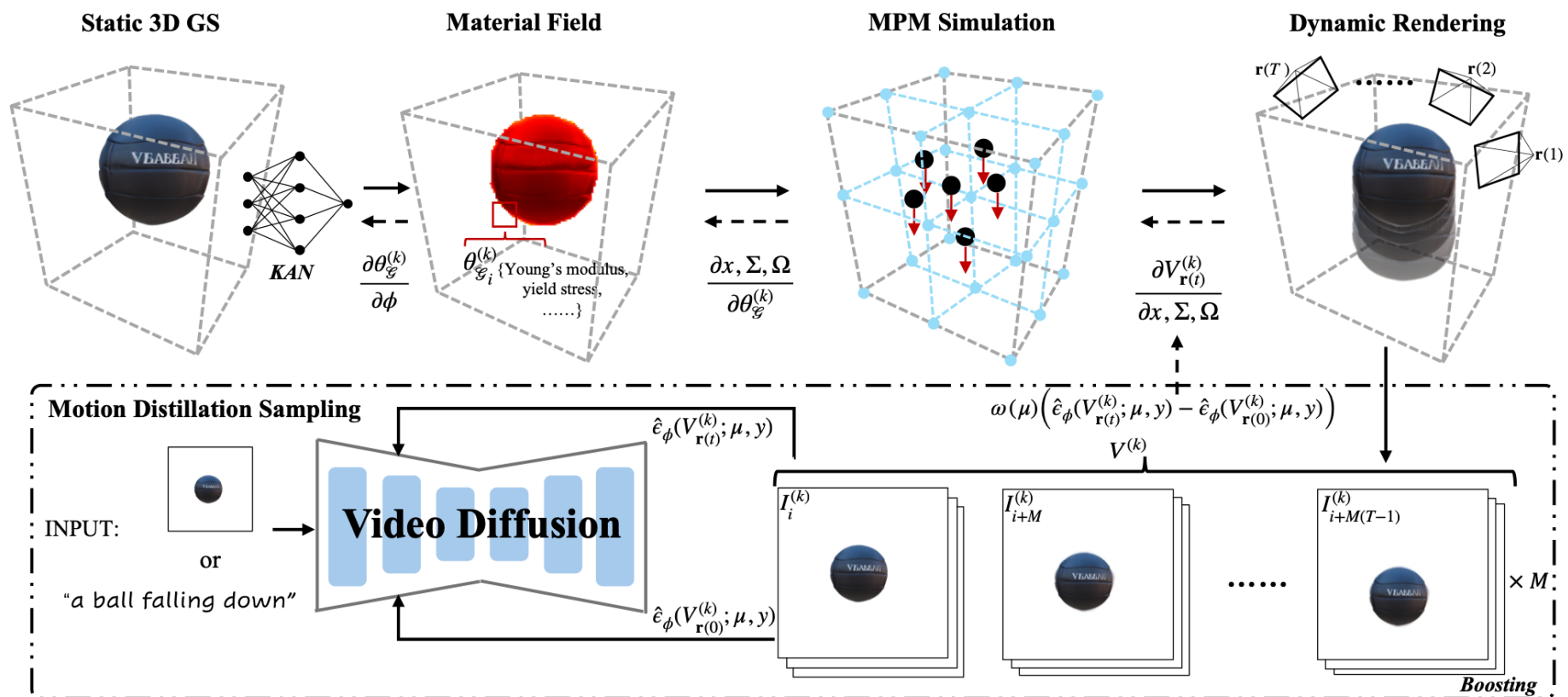
PhysDreamer

- Use a generated reference video to guide physical simulation.



Our DreamPhysics

- Learn material fields with guidance from a video diffusion model via Score Distillation Sampling (SDS)



Results

- Update physical properties to improve the simulated motion.

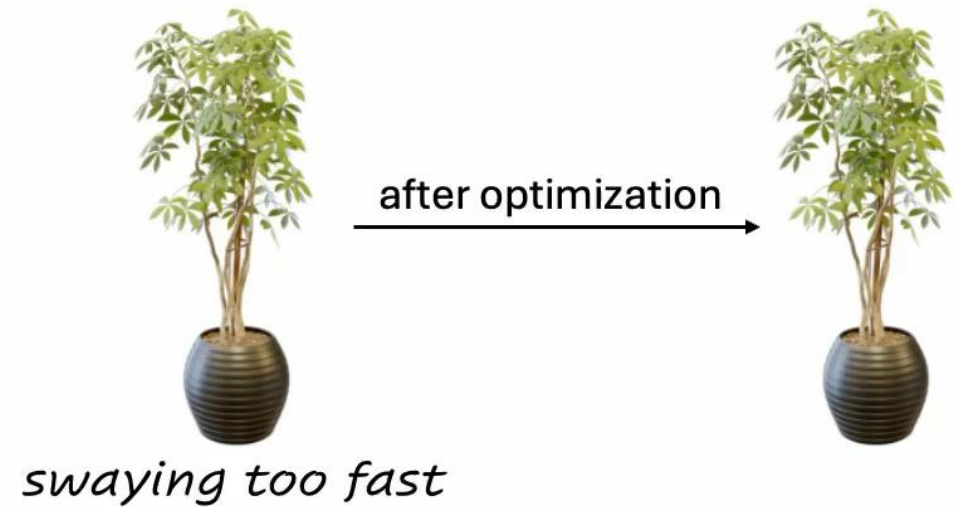


After optimization



Results

- Update physical properties to improve the simulated motion.



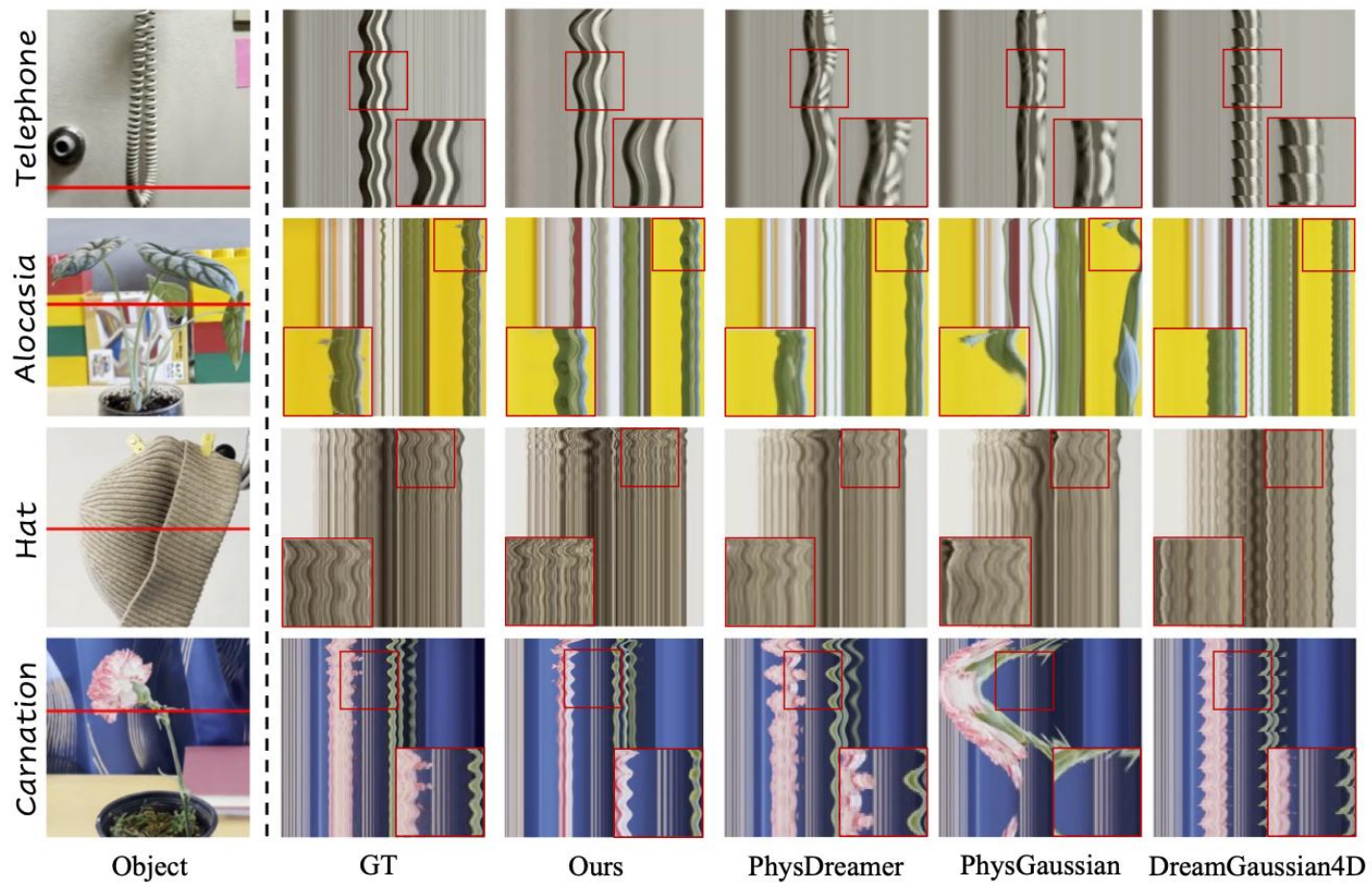
More Results

- Examples of simulated motion for different objects.



Comparison

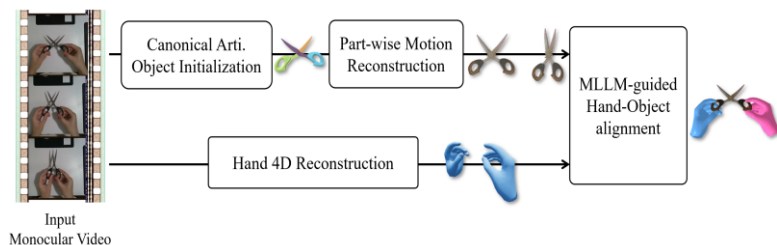
- Visual comparison with previous methods.



Content

ArtHOI (CVPR 2026, Highlight)

Interaction reconstruction

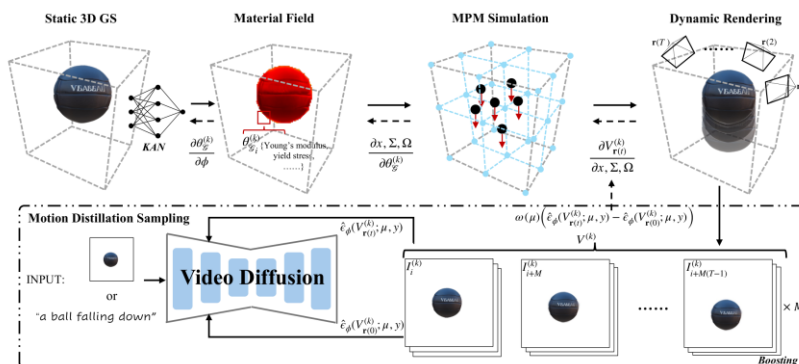


Reconstruct 4D hand-articulated-object interactions

How did the interaction occur?

DreamPhysics (AAAI 2025)

Physical response modeling

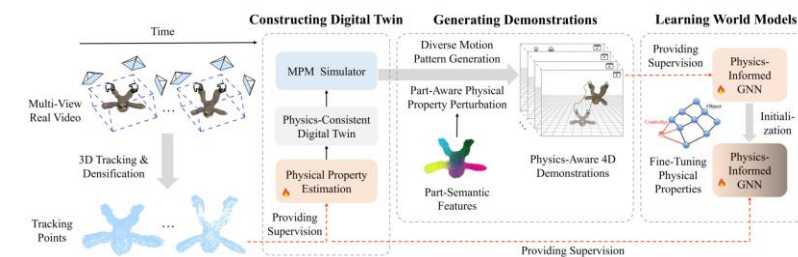


Learn material fields with video diffusion priors

How can the object respond?

PhysWorld (Under Review)

Predictive world modeling



Synthesize physical demonstrations and learn efficient dynamics model of deformable objects

What follows a new action?

World Models of Deformable Objects

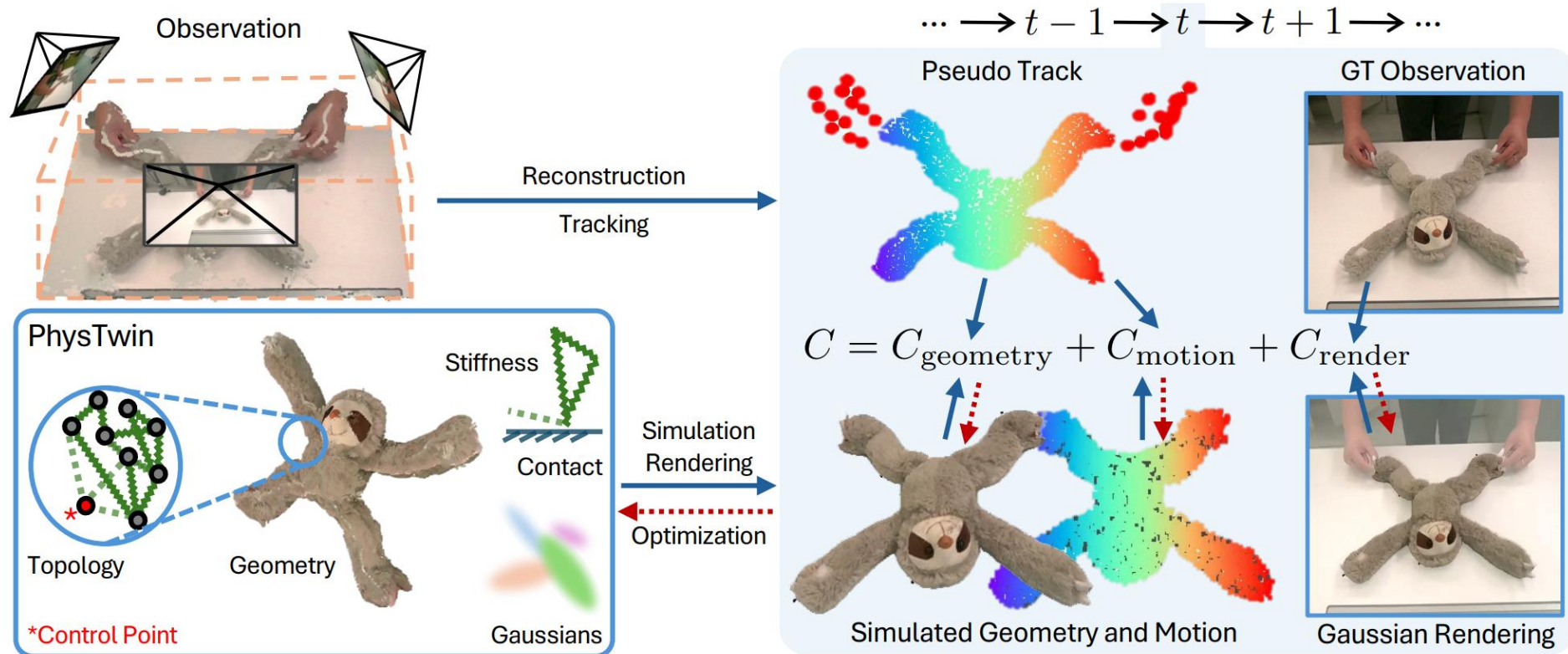
- **World Model:** Predict the next state from the current state and an action.

$$X_{t+1} = f(X_t, a)$$

- **Learning-based models** can make fast predictions, but require large interaction datasets, which are **high cost** to collect in the real world.
- **Physics-based simulators** provide physical priors, but their computational cost limits **real-time prediction**.
- **Our goal** is to combine their strengths: learn an accurate and fast world model from a short real video.

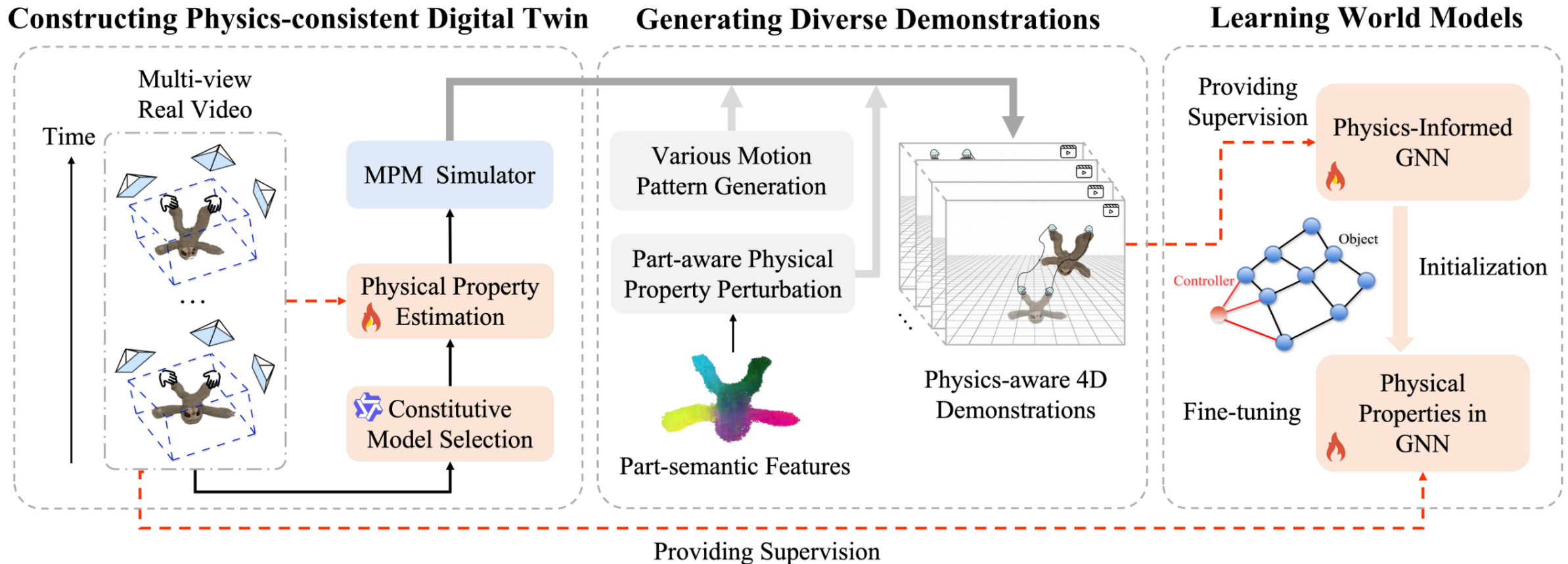
PhysTwin: Spring-Mass Models

- Faster than MPM, but still can't support real-time future state prediction.



Our PhysWorld

- Learning efficient world model from a short real video

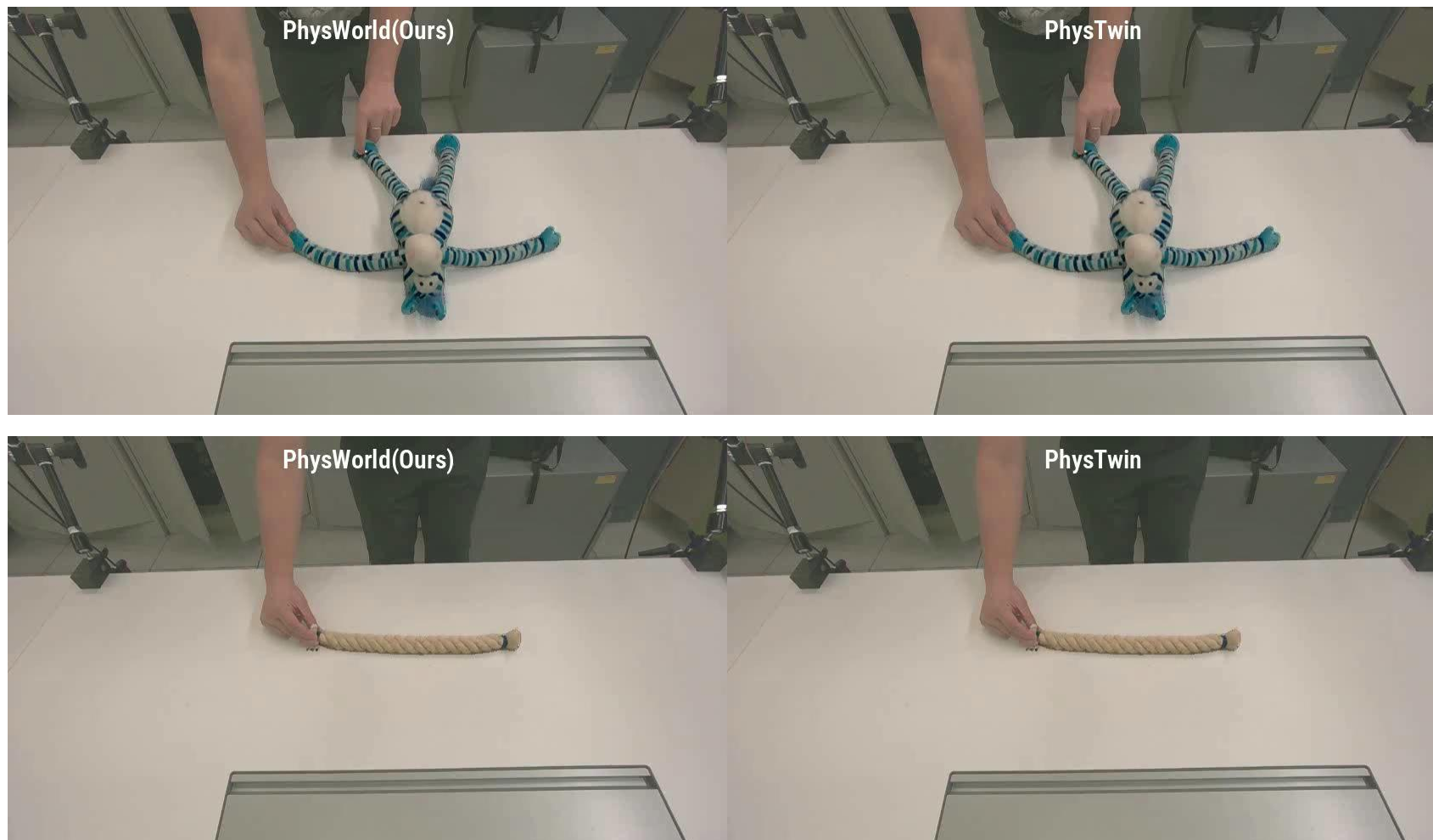


Quantitative Results

- Enable inference speeds 47 times faster than PhysTwin.

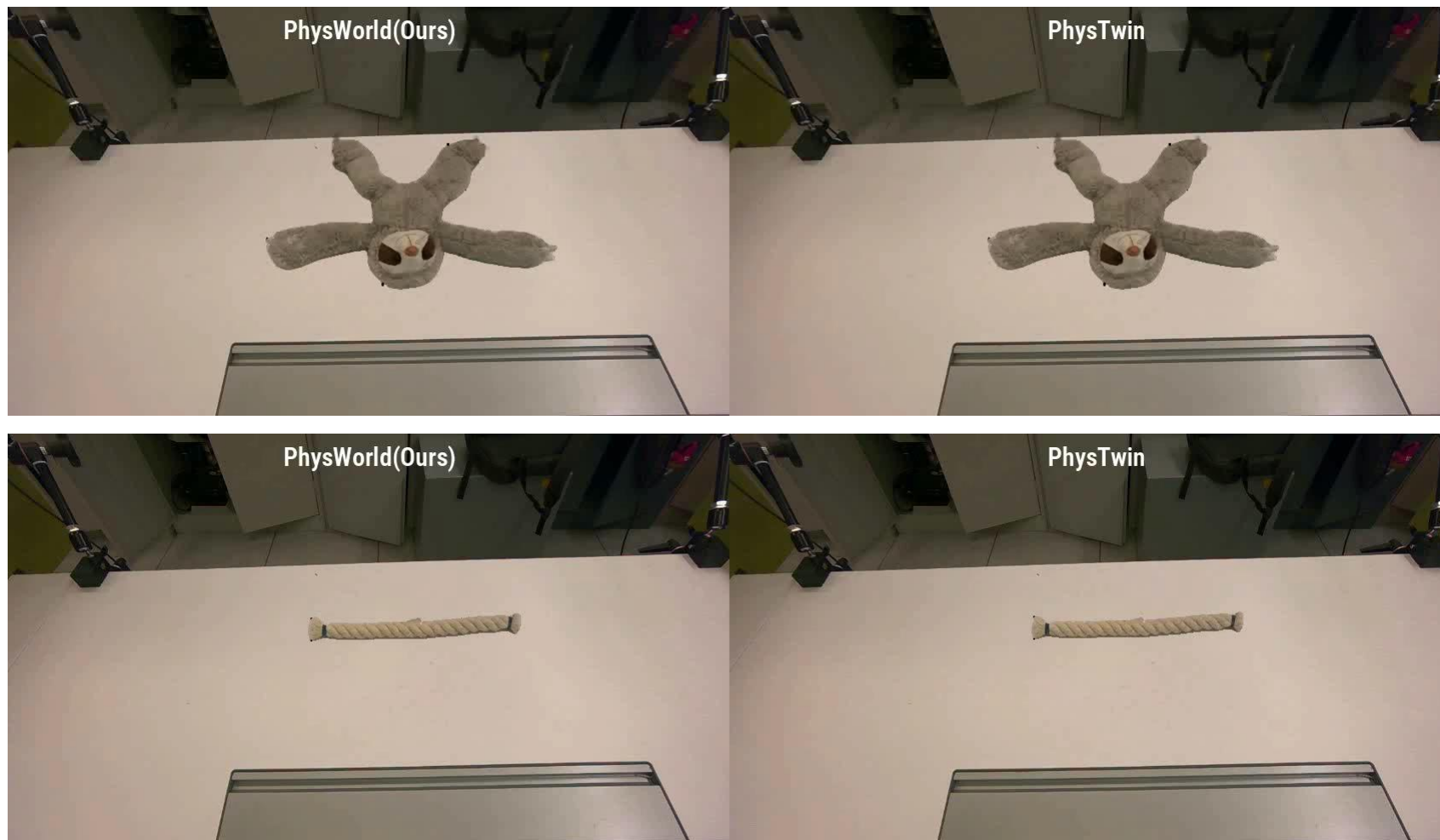
Methods	CD↓	Track↓	IoU↑	PSNR↑	SSIM↑	LPIPS↓	FPS↑
Spring-Gaus (ECCV 2024)	0.062	0.094	46.4	22.488	0.924	0.113	2
GS-Dynamics (CoRL 2024)	0.041	0.070	49.8	22.540	0.924	0.097	<u>236</u>
PhysTwin (ICCV 2025)	0.012	<u>0.022</u>	72.5	25.617	<u>0.941</u>	<u>0.055</u>	17
Our MPM	<u>0.011</u>	0.021	74.7	26.231	0.942	0.052	3
Our GNN w/o Finetuning	0.012	0.025	70.0	25.345	0.939	0.059	799
PhysWorld(Our GNN w Finetuning)	0.010	0.021	<u>73.3</u>	<u>25.940</u>	<u>0.941</u>	<u>0.055</u>	799

Action-Conditioned Future Prediction



The lower-opacity segments represent the ground truth motion of the object.

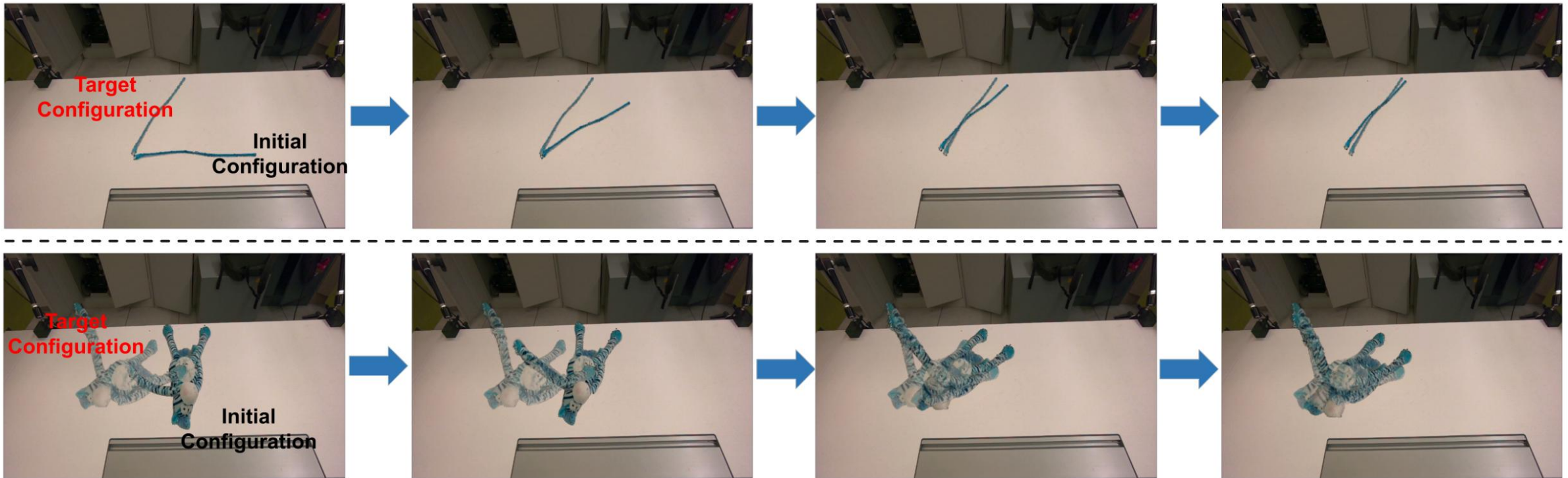
Unseen Interaction



- PhysWorld generates physically plausible predictions,
- PhysTwin suffers from artifacts such as fracture-like rope distortions and unnatural foot folding.

Planning in Simulation Environment

- With MPPI [1] controlling method, the rope (up) and the zebra doll (down) are transferred from the initial configurations to the target ones.



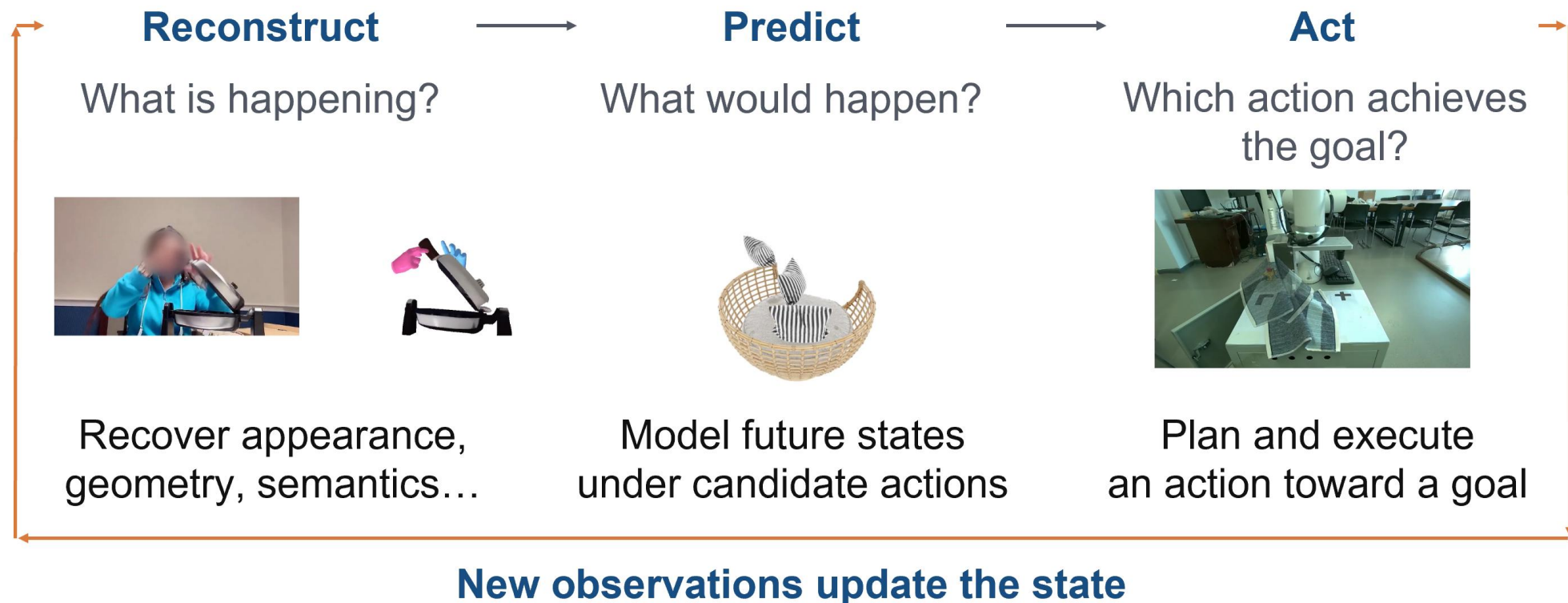
Real Robot Experiments



The semi-transparent object shows the target shape.

Summary

- Learn knowledge from interactions to predict their outcomes and guide actions.
- Build a closed loop of reconstruction, prediction, and action.



Paper

- Zikai Wang, **Zhilu Zhang***, Yiqing Wang, Hui Li, Wangmeng Zuo. ArtHOI: Taming Foundation Models for Monocular 4D Reconstruction of Hand-Articulated-Object Interactions. CVPR 2026, Highlight. [Paper](#) [Code](#) [Project](#)
- Tianyu Huang, Haoze Zhang, Yihan Zeng, **Zhilu Zhang**, Hui Li, Wangmeng Zuo*, Rynson W. H. Lau*. DreamPhysics: Learning Physics-Based 3D Dynamics with Video Diffusion Priors. AAAI 2025. [Paper](#) [Code](#) [Project](#)
- Yu Yang, **Zhilu Zhang***, Xiang Zhang, Yihan Zeng, Hui Li, Wangmeng Zuo. PhysWorld: From Real Videos to World Models of Deformable Objects via Physics-Aware Demonstration Synthesis. Under Review. [Paper](#) [Code](#) [Project](#)

*Corresponding author

Acknowledgments



Professor
Wangmeng Zuo
(左旺孟)



Zikai Wang
(王子恺)



Tianyu Huang
(黄天毓)



Yu Yang
(杨宇)

Thank You

Zhilu Zhang (张志路)

Singapore Management University